

An exact count for OEIS A228464: windowed maxima over a growing alphabet

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Abstract

OEIS A228464 counts the arrays obtainable as the maxima of 3 adjacent elements of a length 9 array with entries in $\{0, \dots, n\}$, and carries an empirical polynomial for that count. The count is not a transfer-matrix count: the length is fixed and it is the ALPHABET that grows with n . It is nevertheless exactly computable, because membership in the image depends only on the relative order of the entries of the candidate array and not on their values, so the count is a fixed non-negative combination of binomial coefficients $\binom{n+1}{m}$. Computing that combination settles the empirical claim as an identity in $\mathbb{Q}[n]$.

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1 The conjecture

The entry's first terms are 44, 383, 1821, 6254, 17487, 42386, 92430, 185727, ... and it states

$$\begin{aligned} \text{Empirical: } a(n) = & (4/315)n^7 + (1/5)n^6 + (91/45)n^5 + (63/8)n^4 + \\ & (2557/180)n^3 + (517/40)n^2 + (2419/420)n + 1 = (n+1) \cdot (n+2) \cdot (32n^5 + \\ & 408n^4 + 3808n^3 + 7605n^2 + 5367n + 1260)/2520. \end{aligned}$$

2 The object

Fix $k = 9$ and $w = 3$, and put $L = k - w + 1 = 7$. For $a = (a_1, \dots, a_k) \in \{0, \dots, n\}^k$ let

$$\Phi(a) = (\max(a_1, \dots, a_w), \max(a_2, \dots, a_{w+1}), \dots, \max(a_{k-w+1}, \dots, a_k)) \in \{0, \dots, n\}^L.$$

The entry counts $|\Phi(\{0, \dots, n\}^k)|$.

3 Membership in the image is decided by the order type

Write $W_j = \{j, j+1, \dots, j+w-1\}$ for the j -th window, $1 \leq j \leq L$.

Lemma 1. *Let $b \in \{0, \dots, n\}^L$ and define $\hat{a} \in \{0, \dots, n\}^k$ by $\hat{a}_i = \min\{b_j : i \in W_j\}$. Then $b \in \Phi(\{0, \dots, n\}^k)$ if and only if $\Phi(\hat{a}) = b$.*

Proof. Each $\hat{a}_i$ is one of the b_j , so $\hat{a}$ does lie in $\{0, \dots, n\}^k$ and the condition is meaningful. If $\Phi(\hat{a}) = b$ there is nothing to prove. Conversely suppose $\Phi(a) = b$ for some a . For every $i \in W_j$ we have $a_i \leq \max_{l \in W_j} a_l = b_j$, so $a_i \leq \min\{b_j : i \in W_j\} = \hat{a}_i$: the array $\hat{a}$ dominates every witness componentwise. Hence $\max_{i \in W_j} \hat{a}_i \geq \max_{i \in W_j} a_i = b_j$. The reverse inequality is immediate from the definition, since $\hat{a}_i \leq b_j$ for each $i \in W_j$. So $\Phi(\hat{a}) = b$. $\square$

Both the construction of $\hat{a}$ and the test $\Phi(\hat{a}) = b$ use only comparisons between entries of b . Consequently:

Corollary 2. *Whether b lies in the image depends only on the order type of b – the weak ordering of its L coordinates – and not on n or on the values themselves.*

Proof. Let σ be any strictly increasing map defined on the values of b . Applying σ entrywise commutes with $\min$ and $\max$, so it carries $\hat{a}$ for b to $\hat{a}$ for $\sigma(b)$ and preserves the equalities tested in Lemma 1. Any two arrays with the same order type differ by such a σ . $\square$

4 The count

For $1 \leq m \leq L$ let A_m be the number of order types of L coordinates that use exactly m distinct values and are achievable. An array $b \in \{0, \dots, n\}^L$ using exactly m distinct values is determined by the m -element subset of $\{0, \dots, n\}$ it uses together with its order type, and every pair occurs exactly once; so by Corollary 2,

$$|\Phi(\{0, \dots, n\}^k)| = \sum_{m=1}^7 A_m \binom{n+1}{m} \quad \text{for every } n \geq 0.$$

Enumerating the weak orderings of 7 elements and applying Lemma 1 to each gives $A_1 = 1, A_2 = 42, A_3 = 254, A_4 = 549, A_5 = 544, A_6 = 272, A_7 = 64$, so

$$|\Phi(\{0, \dots, n\}^k)| = 1 \binom{n+1}{1} + 42 \binom{n+1}{2} + 254 \binom{n+1}{3} + 549 \binom{n+1}{4} + 544 \binom{n+1}{5} + 272 \binom{n+1}{6} + 64 \binom{n+1}{7}$$

5 Conclusion

The right-hand side is the polynomial the entry states, so the empirical claim of OEIS A228464 is true for every $n \geq 0$. The argument is finite and exact: it enumerates the weak orderings of 7 elements, of which there are 1726 achievable, and performs no fitting to the published terms. Those terms are reproduced by the formula as a check, not as evidence.